17. In $\overline{F}[x]$, let $f(x) = (x-\alpha)^\nu g(x)$ where $g(\alpha) \neq 0$ and $\nu \geq 1$ because $f(\alpha) = 0$. Then by Exercise 16, we have

$$f'(x) = (x-\alpha)^\nu g'(x) + \nu(x-\alpha)^{\nu-1} g(x).$$

Remembering that $\nu \geq 1$ and that $g(\alpha) \neq 0$, we see that $f'(\alpha) = 0$ if and only if $\nu > 1$, that is, if and only if α is a zero of $f(x)$ of multiplicity > 1.

18. Let $f(x)$ be an irreducible polynomial in $F[x]$ where F is a field of characteristic 0. Suppose that α is a zero of $f(x)$ in $\overline{F}$. Because $f(x) \in F[x]$ has minimal degree among all nonzero polynomials having α as a zero, we see that $f'(\alpha) \neq 0$, for the degree of $f'(x)$ is always one less than the degree of $f(x)$ in the characteristic 0 case. By Exercise 17, α is a zero of $f(x)$ of multiplicity 1. By Theorem 51.2, all zeros of $f(x)$ have this same multiplicity 1, so $f(x)$ is separable.

19. Let α be a zero of irreducible $q(x)$ in the algebraic closure $\overline{F}$. The argument in Exercise 18 shows that $q'(\alpha) \neq 0$ unless $q'(x)$ should be the zero polynomial. Now $q'(x) = 0$ if and only if each exponent of each term of $q(x)$ is divisible by p. If this is not the case, then $q'(\alpha) \neq 0$ so α has multiplicity 1 by Exercise 17, and so do other zeros of $q(x)$ by Theorem 51.2, so $q(x)$ is a separable polynomial. This proves the "only if" part of the exercise.

Suppose now that every exponent in $q(x)$ is divisible by p. Let $g(x)$ be the polynomial obtained from $q(x)$ by dividing each exponent by p. Then $\alpha \in \overline{F}$ is a zero of $q(x)$ if and only if α^p is a zero of $g(x)$. Let $g(x)$ factor into $(x-\alpha^p)h(x)$ in $\overline{F}[x]$. Then $q(x) = (x^p - \alpha^p)h(x^p) = (x-\alpha)^p h(x^p)$ in $\overline{F}[x]$, showing that α is a zero of $q(x)$ of algebraic multiplicity at least p, so $q(x)$ is not separable.

20. The polynomials $f(x)$ and $f'(x)$ have a common nonconstant factor in $\overline{F}[x]$ if and only if they have a common zero in $\overline{F}$, because a zero of the common nonconstant factor must be a zero of each polynomial, and a common zero α give rise to a common factor $x-\alpha$. Thus by Exercise 17, $f(x)$ and $f'(x)$ having a common factor in $\overline{F}[x]$ is equivalent to $f(x)$ having a zero of multiplicity greater than 1. Therefore there is no nonconstant factor of $f(x)$ and $f'(x)$ in $\overline{F}[x]$ if and only if $f(x)$ has no zero in $\overline{F}$ of multiplicity greater than 1.

21. If $f(x)$ and $f'(x)$ have no nonconstant factor in $\overline{F}[x]$, then they certainly have no nonconstant factor in $F[x]$. Suppose now that they have no nonconstant factor in $F[x]$ so that a gcd of $f(x)$ and $f'(x)$ in $F[x]$ is 1. By Theorem 46.9,

$$1 = h(x)f(x) + g(x)f'(x)$$

for some polynomials $h(x), g(x) \in F[x]$. Viewing this equation in $\overline{F}[x]$, we see that every common factor of $f(x)$ and $f'(x)$ must divide 1, so the only such common factors are elements of $\overline{F}$, and 1 is a gcd of $f(x)$ and $f'(x)$ in $\overline{F}[x]$ also. Thus $f(x)$ and $f'(x)$ have no common nonconstant factor in $F[x]$ if and only if they have no common nonconstant factor in $\overline{F}[x]$. By Exercise 20, this is equivalent to $f(x)$ having no zero in $\overline{F}$ of multiplicity greater than 1.

22. Compute a gcd of $f(x)$ and $f'(x)$ using the Euclidean algorithm. Then $f(x)$ has a zero of multiplicity > 1 in $\overline{F}$ if and only if this gcd is of degree > 0.

52. Totally Inseparable Extensions

1. The separable closure is $\mathbb{Z}_3(y^3, z^9)$ because $(y^3)^4 = u$ and $(z^9)^2 = v$ and 3 does not divide 4 or 2. The field $\mathbb{Z}_3(y,z)$ is clearly totally inseparable over (y^3, z^9).

2. Clearly the separable closure contains y^3. Therefore it must contain $(y^2z^{18})^3/(y^3)^2 = z^{54}$, and hence must contain z^{27}. Because it must also contain y^2z^{18}, we see that the separable closure is $\mathbb{Z}_3(y^3, y^2z^{18}, z^{27})$. Clearly $\mathbb{Z}_3(y, z)$ is totally inseparable over this field.

3. The totally inseparable closure is $\mathbb{Z}_3(y^4, z^2)$.

4. The totally inseparable closure must contain y^4, and therefore $(y^2z^{18})^2/y^4 = z^{36}$, so it must also contain z^4. Of course it must also contain y^2z^{18} and therefore $y^2z^{18}/(z^4)^4 = y^2z^2$. We see the totally inseparable closure is $\mathbb{Z}_3(y^4, y^2z^2, z^4)$. Note that $(y^2z^2)^{27} = (y^{12})^4(y^2z^{18})^3$, and that $(z^4)^{27} = (y^2z^{18})^6/y^{12}$.

5. F T F F F F T F T T

6. If E is a separable extension of F, then there are no elements of E totally inseparable over F, so the totally inseparable closure of F in E is just F, which is a subfield of E.

Suppose that E does contain some elements totally inseparable over F, and let K be the union of F with the set of all such totally inseparable elements. We need only show that for $\alpha, \beta \in K$, the elements $\alpha \pm \beta, \alpha\beta$, and $1/\alpha$ if $\alpha \neq 0$, are either in F or are totally inseparable over F. Suppose that α is not in F, but is totally inseparable over F, so that $\alpha^{p^r} \in F$. Then for any element b in F, we have $(\alpha \pm b)^{p^r} = \alpha^{p^r} \pm b^{p^r}$ and this sum or difference is in F. Also $(b\alpha)^{p^r} = b^{p^r}\alpha^{p^r}$ is in F, and $(1/\alpha)^{p^r} = 1/\alpha^{p^r}$ is in F if $\alpha \neq 0$. This shows that for α in K but not in F and for b in F, the elements $\alpha \pm b, b\alpha$, and $1/\alpha, \alpha \neq 0$, are in K. The other case we have to worry about is where α and β are both totally inseparable over F, that is, they are both in K, but neither one is in F. Then $\alpha^{p^r} \in F$ and $\beta^{p^s} \in F$ for some $r, s \in \mathbb{Z}^+$. Suppose that $s \geq r$. Then $(\alpha \pm \beta)^{p^s} = \alpha^{p^s} \pm \beta^{p^s} = (\alpha^{p^r})^{p^{s-r}} \pm \beta^{p^s}$ is in F, and $(\alpha\beta)^{p^s} = (\alpha^{p^r})^{p^{s-r}}\beta^{p^s}$ is in F. Thus $\alpha \pm \beta$ and $\alpha\beta$ are either already in F or are totally inseparable over F. This shows that K is closed under the field operations of addition, subtraction, multiplication, and contains multiplicative inverses of nonzero elements.

7. Suppose that F is perfect. If $x^p - a$ has no zero in F for some $a \in F$, then $F(\sqrt[p]{a})$ is a proper extension of F and is totally inseparable over F, contradicting the hypothesis that F is perfect. Thus $x^p - a$ has a zero in F for every $a \in F$, that is, $F^p = F$.

Conversely, suppose that $F^p = F$ and let $f(x)$ be an irreducible polynomial in $F[x]$. We must show that $f(x)$ is a separable polynomial. Let E be the separable closure of F in the splitting field K of $f(x)$ in $\overline{F}$. Let $[E : F] = n$. Now the map $\sigma_p : E \to E^p$ is an isomorphism. Because $F^p = F$ and σ_p is one to one, no $\alpha \in E$ that is not in F is carried into F. Because σ_p is an isomorphism, the extension E of degree n over F is carried into an extension E^p of $F^p = F$ of degree n. Because $E^p \leq E$, we see that $n = [E : F] = [E : E^p][E^p : F] = [E : E^p]n$, so $[E : E^p] = 1$ and $E = E^p$. But then E has no totally inseparable extension, for an element α of such an extension must satisfy $\alpha^{p^r} = \beta \in E$, where $\alpha \notin E$. But because $E^p = E$, we see that $E^{p^r} = E$ and the polynomial $x^{p^r} = \beta$ has a zero γ in E, so that $x^{p^r} - \beta = x^{p^r} - \gamma^{p^r} = (x - \gamma)^{p^r}$, showing that γ is the only zero of this polynomial. Thus no such $\alpha \notin E$ exists and we must have $E = K$, so the splitting field of $f(x)$ is a separable extension of F, and therefore $f(x)$ is a separable polynomial.

8. The solution of Exercise 7 showed that if $F^p = F$, then $E^p = E$. Conversely, suppose that $E^p = E$. Let $n = [E : F]$. Because σ_p is an isomorphism, it must be that $[E^p : F^p] = n$. Of course $F^p \leq F$. Then we have

$$n = [E^p : F^p] = [E : F^p] = [E : F][F : F^p] = n[F : F^p],$$

so $[F : F^p] = 1$ and $F = F^p$.